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The Electoral Cost of the Prime Ministerial Party in OECD Democracies

Is there an incumbency advantage for politicians in
modern democratic elections, and what are the main
reasons behind it?

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Abstract

This paper tests whether prime ministerial (PM) parties face a different electoral cost during their incumbency than other governing parties. The reference all-party benchmark is the one introduced by Nannestad and Paldam (2000, p.1). Nannestad and Paldam observe an average cost of ruling of 2.25 percentage points (pp) each ruling term. This paper employs an empirical design similar to that of the Nannestad and Paldam benchmark, calculating $C_e = V_e - L_{e-1}$ for the incumbent PM party at each re-election bid, including re-elections where the party lost power (Nannestad & Paldam, 2000). This approach ensures comparability to the existing benchmark effect. The analysis draws on the ParlGov Dataset (Döring & Manow, 2024) and analyses 102 prime ministerial incumbency elections across 20 OECD countries from 1990 to 2020, adhering to the overall topic of modern democratic elections. The central result is a mean electoral penalty for PM incumbents of -3.80 pp per ruling term ($SD = 6.12$, $SE = 0.61$). This cost is significantly negative in direction ($t = -6.28$, $p < 0.001$, one-sided) as well as significantly larger in magnitude than the benchmark effect of -2.25 pp ($t = -2.56$, $p = 0.006$, one-sided). The pattern is also consistent geographically: 19 of the 20 countries analysed show a negative average PM electoral cost. These results reject the idea that PM parties are insulated from the cost of ruling through high visibility and name-recognition. PM parties in the analysed sample face a larger penalty than the all-party average, which can be explained through the theory of concentrated electoral accountability.

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Abbreviation	Definition
C_e	Election outcome for the incumbent party
CT	Consecutive Terms counter
H1/H2/H3	Research hypotheses 1, 2, and 3
L_{e-1}	Vote share of the same party at the previous election (e-1).....
N&P	Nannestad and Paldam
n.s.	Not Significant
OECD	Organisation for Economic Co-operation and Development
p	Probability value
ParlGov	Parliaments and Governments (political data infrastructure)
PM	Prime Minister
pp	Percentage Points
SD	Standard Deviation
SE	Standard Error
t	t-value
V_e	Vote share of the incumbent PM party at election e
vs.	Versus

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1 Introduction

Governing parties lose votes. Empirically stable across different democratic systems and time periods, parties in power tend to lose votes in subsequent elections. This observation is so consistent that Paldam (1986, p.23) labels it as a general law in electoral politics. Nannestad and Paldam (2000, p.1) expanded this into a popular and often cited benchmark. They find that on average, the governing party loses about 2.25 percentage points (pp) of its vote share per ruling term. Their sample consisted of 282 democratic elections from 1948 to 1997. This number became a reference point for subsequent modern research on incumbency effects and therefore also serves as a benchmark measure for the paper at hand.

The research of Nannestad and Paldam (2000) (N&P from now on) does not discriminate between positions and treats all governing parties as equivalent, including prime ministerial (PM) parties, junior coalition partners and others. The benchmark does not address whether there is a difference in electoral effect between incumbent positions. Do parties holding the PM seat face the same or different costs than the mean of all governing parties? This question allows for two opposing predictions. On one side, PM parties hold the most visible and accountable position in government. PM parties are the clearest target for voter punishment, drawing on the clarity of responsibility framework of Powell and Whitten (1993, p.398). This would lead to the prediction that PM ruling-costs are more negative than the all-party average. On the other side, PM parties have executive control in steering the government agenda and could benefit from the ability to frame their actions, influencing the perception of their performance (Bélanger & Nadeau, 2015, pp.909-910). This could partially counter the costs. The question is, which force is stronger. The specific research question of this paper is: Do prime ministerial parties in OECD democracies face a different electoral incumbency effect than the all-party cost-of-ruling benchmark?

In order to answer the question at hand, this paper needs a measurement design symmetrical to the N&P benchmark. Therefore, the change in vote share is computed as follows: $C_e = V_e - L_{e-1}$. V_e is the vote share at the more recent election and L_{e-1} the vote share at the preceding election. C_e denotes the change in the incumbent PM party's vote share between one election and the subsequent, for each PM re-election. This measurement method includes elections where the incumbent PM party subsequently lost power, which was falsely not included in a previous version of the paper at hand. As a consequent learning: Incumbency measurements that are restricted to elections where the incumbent stays in office lead to an upward-biased estimate of the PM incumbency cost.

By applying a N&P symmetric design to a sample of 20 OECD democracies between 1990 and 2020, treating 102 prime ministerial incumbent elections, this paper finds an average PM electoral cost of -3.80 pp. This result is significant, negative in direction and significantly more negative than the all-party N&P benchmark of -2.25 pp. It rejects the idea that PM parties are shielded from electoral effects. PM parties actually face a higher cost of ruling compared to the all-party average. This result is also consistent geographically, with 19 out of 20 sample countries showing a negative mean PM electoral cost.

The findings are consistent with the theory of concentrated accountability (Powell & Whitten, 1993, pp.398-400). Since voters can easily and unambiguously identify the Prime Minister's party as primary government actor, electoral punishment is larger and more direct for incumbent prime ministerial parties than for junior coalition partners. Non-PM parties can probably distance themselves more credibly from implemented government policy and its (negative) effects. The result also suggests that visibility premiums cannot offset the electoral cost of holding the most powerful office.

The remainder of this paper is structured as follows: Section 2 encompasses a theoretical literature review, treating mechanisms behind the cost of ruling and the mechanisms that could explain effect differences between PM and non-PM governing parties. Section 3 describes the data used, the sample underlying empirical findings and the design of measurement. Section 4 treats the main results. Section 5 discusses these findings and their implications, also including important limitations. Section 6 concludes.

2 Theoretical framework

2.1 The cost of ruling

In political literature the penalty on incumbency is a robust finding. Governing parties tend to lose vote share when ruling. Among the first to document a "cost of ruling" was Paldam (1986, pp.5-6). He found that across his seventeen observed democracies, incumbent parties consistently lose voters during their rule. Later, Nannestad and Paldam (2000, p.1) analysed the effect on a broader sample, observing 282 democratic elections from 1948 through 1997. They found an average cost of -2.25 pp on vote share per ruling term. They find that the effect is stable across countries and institutional design. This effect is further developed by Stevenson (2002, p.157). He finds that there is a time component of incumbency effects. The longer a party remains in office, the higher the penalty on incumbency.

Nannestad and Paldam (2000, pp.10-11) propose three mechanisms for explaining the burden on incumbency. The first theory encompasses the coalition of minorities. Every decision that a ruling government implements creates a small subgroup of discontent voters who turn against the ruling government. Over time, discontent voters accumulate and at some point form a coalition large enough to unseat the incumbent. The second of the three theories is the median gap theory based on Hotelling's (1929, pp.54-57) theory of spatial competition. The theory assumes a simplified one-dimensional party space from left to right. The median voter has position M ($L - M - R$). In campaigns, parties converge their positions towards the median voters preference to maximize the number of supporters. In office, parties circle back towards their original ideologies. When observing the discrepancy, the median voter shifts back to the opposing party to bring policy back to the centre M . This theory seems abstract, which Nannestad and Paldam (2000, pp.11-12) acknowledge. Nonetheless, Wlezien (2017, pp.730-731) finds a positive correlation between divergence from the promised policy and the cost of ruling. The third explanation for the cost of ruling is the grievance asymmetry theory. It is based on a psychological negativity bias and loss aversion complex, as treated by Kahneman and Tversky (1979, p.263). The core mechanism of the theory is that voters respond asymmetrically to positive and negative policy outcomes. Negative developments are weighted more heavily than positive developments of the same magnitude. This implies that even an incumbent performing well can suffer from the cost of ruling due to a psychological amplification of negative developments. Nannestad and Paldam (2000, pp.13-14) state this theory as the most promising explanation for the incumbency penalty. The psychological component supports the stability of ruling-costs between countries and institutional designs.

Thesen et al. (2024, p.1601) further introduce a media-perspective on the cost of ruling.

They argue that the incumbent faces a disproportionately negative news coverage. This is rooted in the media’s tendency to be critical towards government power. Over time, the negative coverage translates to fewer votes, contributing to the cost of ruling.

The mechanisms underlying the cost of ruling can be divided into two broad categories. The first category is retrospective economic voting, where voters evaluate government performance based on economic outcomes and reward or punish incumbents accordingly (Lewis-Beck, 1990, pp.3-8). Bad economic developments lead to a loss in vote share, while good developments enhance the incumbents standing. The second mechanism is a structural protest effect. Even in times of favourable economic circumstances, voters use each election as an opportunity to signal dissatisfaction with the incumbent government. This produces a consistent negative penalty independent of actual performance (Paldam, 1986, pp.5-10). Both mechanisms predict $C_e < 0$ for governing parties, but neither predicts whether PM parties will face a larger or smaller cost than average.

An important note is that the cost of ruling should be treated separately from the political business cycle. The latter refers to the deliberate manipulation of economic policy by incumbents ahead of elections, in order to improve their electoral prospects (Nordhaus, 1975, pp.169-182). This paper does not incorporate macroeconomic controls and political business cycles and acknowledges this as a limitation. However, the magnitude and consistency of the estimated PM cost of -3.80 pp across different institutional contexts suggests that macroeconomic manipulation is unlikely to explain the result.

2.2 The incumbency advantage

The opposite incumbency effect is documented in the literature on incumbency advantage mechanisms, heavily rooted in US presidential elections. It is nevertheless relevant to the present paper to understand possible differences between standardized all-party cost of ruling concepts and the analysed PM incumbency effects. Mayhew provides an important theoretical framework to understand incumbency advantages. He argues that the primary goal of U.S. congress members is re-election. Following from this, legislators focus their resources on advertising, credit claiming, and position taking, aimed at strengthening their own position (Dodd, 1978, p.693). According to this theory, officeholders actively exploit incumbency advantages through strategic behaviour. Lee (2008, pp.466-467) empirically tested the incumbency effect by applying a regression discontinuity design to obtain near-experimental causal estimates of U.S congressional incumbency advantages. He found a vote share boost of approximately 8 percentage points, increasing seat retention probability by 40–45 percentage points.

Theoretical explanations for incumbency advantages can be grouped into two categories. The first category encompasses theories with a focus on the individual incumbents and their activities. Examples include name recognition, constituency service and office experience. The second category encompasses theories regarding partisan mechanisms, referring to the benefit that a non-incumbent gains by running for the incumbent party. Fowler and Hall (2014, pp.501-531) tested the personal and partisan components of the overall incumbency effect. They found that there is a substantial personal effect, while the partisan effect is statistically indistinguishable from zero. This finding suggests that U.S. incumbency effects are strong in enhancing the individuals success and not statistically relevant for improving party success independent of the incumbent president.

In modern parliamentary systems, there is little evidence of substantial incumbency advantages. There are findings of positive incumbency effects in Canada with district based electoral elements (Kendall & Rekkas, 2012, pp.1560-1561). However, the international evidence generally supports a cost of ruling framework over incumbency advantages (Nannestad & Paldam, 2000, p.1).

Generally, the incumbency advantage framework offers limited application to parliamentary systems, which are the subject of the paper at hand. However, it does allow for the theoretical assumption that prime ministerial parties, whose leader is the most visible political figure in the country, might benefit from a leadership premium, an offset to the structural cost of ruling stemming from the personal popularity of a popular prime minister. Whether this premium is large enough to substantially reduce the PM party's cost of ruling is the question to be answered.

2.3 Valence politics and leader effects

Stokes (1963, p.373) defined the difference between position issues and valence issues as follows: In position issues parties compete on opposing views whereas in valence issues parties compete on competence rather than direction. Research on valence politics suggests that voters decide not only based on the direction of the party but also on perceived competence of a party. Strengthening the importance of perceived competence, Clarke et al. (2004, p.326) find that the prime ministerial approval is among the most important predictors of voting decisions. Additionally, Bélanger and Nadeau (2015, pp.909-910) demonstrate that leader image is highly relevant for electoral outcomes. These mechanisms support that a popular prime minister can generate a personalised vote premium that could partially offset the cost of ruling.

Conversely, when a prime minister is unpopular or the government's record is poor, the accountability of the PM's party is amplified in the opposite direction. Therefore, the net

effect of the leadership premium on the average cost to the PM party is a question that cannot be resolved from theory alone.

2.4 Clarity of responsibility

If voters engage in economic voting, they must be able to attribute blame or praise to parties. Whether voters hold the correct party accountable for economic developments depends on the clarity of responsibility. Powell and Whitten (1993, p.398) formed the clarity of responsibility framework. If institutional design allows for transparent policy attribution, voters engage more heavily in economic voting. Hobolt et al. (2013, p.179) expand on this approach, finding that government clarity is more relevant than institutional clarity regarding performance voting. Institutional clarity in this sense denotes how transparent the formal power distribution is, whereas government clarity is about how transparent informal decision-making power is. A one-person government allows for higher government clarity than a coalition.

In the context of PM parties, Duch and Falcó-Gimeno (2022, p.757) find that voters attribute blame disproportionately to the PM incumbent, because it is perceived as the party with the most agenda setting power (high government clarity). These findings allow for the assumption that the economic vote is heavily focused on PM parties. The effect of this is twofold: The PM party receives more praise in economic boom and stronger backlash in periods of low performance.

2.5 Hypothesis design

The literature review leads to the three following hypotheses:

H1: The mean electoral cost for incumbent PM parties in re-elections is negative

Rationale: The PM's party holds the most visible governing office and is subject to all the cost of ruling mechanisms. The hypothesis assumes that the discussed possible PM benefits cannot counter structural incumbency disadvantages.

H2: The PM party's incumbency cost is more negative than the all-party benchmark of -2.25 pp.

Rationale: This follows from the clarity-of-responsibility argument. PM parties are held accountable as the primary, most visible target of attribution within coalition governments. Junior partners can hide while PM parties cannot.

H3: The PM incumbency cost is directionally consistent across a majority of sample countries.

Rationale: The effect is not dependent on institutional design but a key feature of holding the PM seat. If the benchmark effect of -2.25 pp is geographically consistent, PM specific effects should be too.

3 Data

3.1 Data sources

The empirical analysis draws on the Parliaments and Governments (ParlGov) Database (Döring & Manow, 2024). This is a comprehensive dataset covering elections and cabinet formations for the countries relevant for the analysed sample and far beyond. The data ranges back to the early twentieth century. The database is maintained by Holger Döring and Philip Manow. For all national elections, ParlGov tracks vote shares of competing parties and attaches a key to each party, allowing parties to be tracked across elections and time. The relevant information is held in two linked tables: "view_election" (one row per party election) and "view_cabinet" (one row per party cabinet), which are linked by "election_date". The party of the Prime Minister at each election is identified through the binary "prime_minister" indicator in the cabinet table. All observations in this paper are drawn from the dataset of August 2024, fully covering the analysed time frame from 1990 to 2020. All 20 countries that are part of the sample are included in the dataset. No additional data sources are used concerning the main hypothesis testing.

This single source approach does not include macroeconomic control variables, which is a key limitation of this analysis. The incorporation of macroeconomic circumstances would be a valuable extension to the following analysis.

3.2 Sample

The analysis of this paper covers 20 OECD member states that have been stable members throughout 1990–2020. Namely, those are: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Japan, Luxembourg, the Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, and the United Kingdom. The selection of these states out of the full OECD-range is mainly due to many OECD-members joining later than 1990. Further, Turkey has been excluded due to doubts regarding competitive democratic processes, the US is excluded due to its presidential system. Switzerland is not part of the sample because it has no Prime Minister in the conventional sense. Italy provides no incumbency results in the relevant time frame. The case of Italy will be discussed further in section 5.2, because it does in theory fit the PM design and the OECD sample criteria. The selection of countries fits the overarching topic of modern democracies and is sufficiently large to produce robust results.

The sample period is intentionally chosen, beginning in 1990 and excluding the cold war period with different political circumstances. The period ends in 2020 to exclude exceptional conditions during the COVID-19 pandemic, which could have influenced electoral

outcomes. Only parliamentary elections are part of the sample, excluding presidential elections and referendums.

To track incumbency, the "CT counter" tracks consecutive terms for each PM party: $CT = 1$ denotes the first election at which a given party holds the PM office, $CT = 2$ denotes the first re-election after serving a full term, $CT \geq 3$ covers subsequent re-elections. Only observations with $CT \geq 2$ are part of the analysis: $CT = 1$ is an entry-to-power event and not yet an incumbent party. The sample excludes 73 $CT = 1$ observations within the 1990–2020 window.

Italy has no $CT \geq 2$ observations in the sample. ParlGov records every Italian election between 1990 and 2020 as a new party taking the PM office ($CT = 1$). This is an artefact of extreme volatility in the Italian party system. Without Italy, the number of countries observed in the sample is $N = 20$. The final sample contains 102 PM incumbency elections across 20 countries with at least one $CT \geq 2$ observation.

The distribution of observations across countries shows actual variation in how often PM parties seek and succeed in re-election. Countries with single-party governments or stable coalitions, such as Australia, Austria, the United Kingdom and New Zealand, tend to produce more $CT \geq 2$ observations because the PM party frequently remains in power over multiple elections. Inversely, countries with highly fragmented or volatile party systems, such as Italy and Greece after 2012, produce fewer (or no) observations because the PM party frequently changes. This structural variation is not a selection problem for the present paper, as the analysis is based on elections where the CT counter reaches at least 2. However, it does not address the prior question of how often a PM party reaches that threshold. The question addressed is: when a PM party does face a re-election, how much does it lose on average?

3.3 Measurement design

The incumbency effect variable C_e is defined exactly as in Nannestad and Paldam (2000, p.2):

$$C_e = V_e - L_{e-1} \tag{1}$$

,where V_e is the vote share of the party in power (PM) at election e , and L_{e-1} is the same party's vote share at the election held at $e - 1$, immediately preceding election e . Therefore, C_e is the difference in the PM party's vote share between two directly consecutive elections in which it held the PM office. This formula is included because it is utilised symmetrically by Nannestad and Paldam (2000). A more intuitive form of the

same equation would be the following:

$$\Delta V = V_t - V_{t-1} \quad (2)$$

,with ΔV denoting the vote share change between the more recent election V_t and the preceding election V_{t-1} .

For the CT counter, the incumbent at election e is defined as the party that held the prime minister role after the previous election ($e-1$). The CT counter rises by one every time the same party wins in re-election. When a party fails in re-election, CT is set back to one. In ParlGov, multiple mid-term cabinet reshuffling entries after a single election are handled by taking the initial PM put forward after each election as the reference government, which is consistent with how N&P define the pre-election government (Nannestad & Paldam, 2000).

3.4 Strategy

The primary analysis method used is the one-sample t-test, which tests whether the mean C_e for the 102 PM incumbency elections differs significantly from its respective null hypothesis value. For H1, the null hypothesis is that the incumbency effect is zero, and the alternative hypothesis is that the cost of incumbency is below zero. For H2, the null hypothesis is -2.25 pp (the N&P benchmark), and the alternative hypothesis is one-sided, suggesting the observed value is lower than -2.25 pp ($C_e < -2.25$ pp). H1 and H2 are directional hypotheses stemming from the theory in Section 2. Because we cannot completely rule out an effect in the opposite direction, two-sided values are also computed. Both one-sided and two-sided p values lead to identical conclusions at $\alpha = 0.05$. Reporting both values shows that the results are not dependent on the directional assumptions. A one-sided binomial test is used for H3, examining whether more than half of the 20 countries show a negative mean PM cost. A significance threshold of $\alpha = 0.05$ is applied throughout.

C_e is an interval-scale measurement with a large underlying sample ($N = 102$). The central limit theorem supports that the sampling distribution of the mean is approximately normal, even if the distribution of individual C_e values is not exactly normally distributed (Shapiro–Wilk $W = 0.950$, $p = 0.001$; negative skewness = -0.95; excess kurtosis = 1.72). Bootstrap confidence intervals (10,000 replications, seed = 42) further support that the parametric (normal distribution) and non-parametric (non-normal distribution) intervals are almost identical. One limitation of the t-test is that it treats all 102 observations as independent. With different institutional designs, observations are not fully independent within countries.

4 Results

Note: The python replication code, encompassing all statistical results, can be found in the appendix. From now on, all data discussed is a product of the ParlGov dataset (Döring & Manow, 2024).

4.1 Descriptive statistics

Table 1 displays the descriptive statistics covering the 102 PM incumbency ($CT \geq 2$) elections in the sample. The mean C_e is -3.80 pp, with a standard deviation of 6.12 pp and a standard error of 0.61 pp. The median C_e is -3.28 pp, which is close to the mean and supports that the overall effect is not driven by outliers. The min-max range spans from -26.67 pp (Canada, 1993, when the Progressive Conservatives collapsed from 42.7% to 16.0%) to +7.70 percentage points (Germany, 2013: the CDU/CSU rebounded from 33.8% to 41.5%) (Döring & Manow, 2024).

The distribution is left-skewed (skewness = -0.95) with moderate excess kurtosis (1.72). This is an artefact of some large negative observations denoting elections in which the PM parties suffered substantial collapses. These are real events that should be included in the sample because they do represent the full cost of incumbency and not measurement errors. Excluding the two most extreme observations (Canada in 1993 and Ireland in 2011) shifts the mean to -3.42 percentage points. This is still well below the N&P benchmark.

Table 1: Descriptive Statistics of PM Incumbency Elections, 1990–2020

Statistic	All PM ($CT \geq 2$)	CT = 2 (1st re-election)	CT ≥ 3 (2nd+ re-election)
N	102	44	58
Mean (pp)	-3.80	-3.68	-3.89
Median (pp)	-3.28	-3.73	-3.25
SD (pp)	6.12	6.45	5.90
SE (pp)	0.61	0.97	0.78
Min (pp)	-26.67	-26.67	-24.11
Max (pp)	+7.70	+7.70	+6.09
N&P benchmark (pp)	-2.25	—	—

Note: CT = Consecutive Terms. Nannestad and Paldam (2000, p.1). Data Source: ParlGov (Döring & Manow, 2024).

Table 2: Hypothesis Test Results

Hypothesis	Test	Null value	Test statistic
H1: Mean $C_e < 0$	One-sample t-test	0 pp	$t = -6.28$ $p < 0.001$
H2: Mean $C_e < -2.25$ pp	One-sample t-test	-2.25 pp	$t = -2.56$ $p = 0.006$
H3: >50% countries neg.	Binomial test	$\pi = 0.50$	19/20 countries $p < 0.001$

Note: H1 alternative: $\mu < 0$. H2 alternative: $\mu < -2.25$ pp. H1/H2 one-sided, two-sided p-values reach identical conclusions. H3 alternative: $\pi > 0.50$ (one-sided binomial). t-tests: $df = 101$. $N = 102$ PM incumbency elections, 1990–2020. $\alpha = 0.05$.

4.2 H1: Is the PM incumbency effect negative?

H1 predicts a sub-zero incumbency effect (C_e) for PM parties.

A one-sample t-test against zero gives the following result: $t(101) = -6.28, p < 0.001$ (one-sided; $p < 0.001$, two-sided). The parametric 95% confidence interval is [-5.00 pp, -2.60 pp], with the corresponding bootstrap 95% confidence interval being [-5.02 pp, -2.63 pp]. Both the parametric and non-parametric intervals exclude zero by a large enough margin. **H0 is rejected at the 5% level: PM parties face an incumbency cost that is negative in direction, relevant in magnitude and statistically significant.**

The cost is observable in both incumbency-term categories, at first re-elections and also at subsequent re-elections. The first re-elections ($CT = 2 : -3.68\text{pp}, N = 44, t = -3.78, p < 0.001$) and subsequent re-election results ($CT \geq 3 : -3.89\text{pp}, N = 58, t = -5.02, p < 0.001$) are nearly identical in magnitude. This finding suggests that the cost does not substantially increase with longer incumbency within the 1990–2020 period. This will be further discussed in section 4.6.

4.3 H2: Is the PM cost larger than the benchmark?

H2 predicts that the mean PM C_e is more negative in magnitude than the N&P all-party benchmark of -2.25 pp.

A one-sample t-test against the benchmark value of -2.25 yields $t(101) = -2.56, p = 0.006$ (one-sided; $p = 0.012$, two-sided). The mean PM incumbency effect of -3.80 pp lies 1.55 pp below the benchmark. **H0 is rejected at the 5% level: PM parties face a substantially larger cost of ruling than the all-party benchmark. The difference is statistically significant.**

This comparison is methodologically feasible because the two measurement designs are

deliberately symmetrical. Both calculate $C_e = V_e - L_{e-1}$ for the incumbent government party at every re-election it faces, regardless of winning or losing said re-election. The main difference, which is also this paper’s contribution, is that N&P treat all cabinet parties equivalently, whereas this paper only observes the prime ministerial party. The larger negative effect experienced by PM parties of 1.55 percentage points below the benchmark therefore represents the surplus cost of holding the top government office. This finding is consistent with the clarity of responsibility mechanism discussed in Section 2.4.

4.4 H3: Geographic consistency

H3 predicts that a majority of sample countries show a negative mean C_e .

19 of 20 countries with $CT \geq 2$ observations show a negative mean PM incumbency effect (95%). The only exception is Luxembourg (+0.40 pp, $N = 5$). In Luxembourg the Christian Social People’s Party has been a particularly stable incumbent. A one-sided binomial test of $H_0: \pi = 0.5$ against $H_1: \pi > 0.5$ gives $p < 0.001$. **H0 is rejected at the 5% level: The negative PM incumbency cost is an almost universal observation across the OECD sample.**

Table 3 displays country-level results. The magnitude of the incumbency cost varies considerably. Starting from -12.42 percentage points in France ($N = 1$, semi-presidential context; borderline result, cautious in interpretation) to +0.40 pp in Luxembourg, the direction never reaches large positive magnitudes with only one above zero mean. Countries with proportional representation and frequent coalition governments tend to show larger PM costs. This is consistent with the prediction that concentrated accountability is particularly strong when coalition design makes it easy for voters to blame (or reward) the PM party for government outcomes.

Table 3: Country-Level Mean PM Electoral Cost (N&P-Comparable Design, 1990–2020)

Country	N	Won	Lost	Mean C_e (pp)	Median (pp)	SD	Direction
[†] France	1	0	1	−12.42	−12.42	—	Negative
Netherlands	5	2	3	−9.45	−12.90	5.38	Negative
Portugal	5	1	4	−8.42	−8.49	6.22	Negative
Canada	6	3	3	−6.79	−5.31	10.61	Negative
Spain	6	2	4	−6.00	−4.02	8.36	Negative
Ireland	6	5	1	−5.23	−2.29	9.69	Negative
Norway	3	0	3	−5.17	−5.20	0.55	Negative
New Zealand	7	3	4	−3.94	−2.64	4.59	Negative
Sweden	4	1	3	−3.35	−5.10	4.66	Negative
Greece	4	1	3	−3.31	−3.59	4.38	Negative
Iceland	5	4	1	−2.98	−1.50	6.98	Negative
Finland	2	0	2	−2.85	−2.85	6.29	Negative
United Kingdom	6	4	2	−2.77	−2.95	6.00	Negative
Denmark	5	2	3	−2.68	−2.80	2.75	Negative
Belgium	5	3	2	−2.42	−3.11	1.61	Negative
Germany	6	4	2	−2.38	−3.30	5.70	Negative
Japan	6	4	2	−2.37	−1.55	7.13	Negative
Austria	7	5	2	−1.28	−0.30	4.13	Negative
Australia	8	5	3	−1.24	−2.44	4.74	Negative
Luxembourg	5	4	1	+0.40	+0.01	4.00	Pos.
Italy	0	—	—	—	—	—	No obs.
Total/Mean	102	53	49	−3.80	−3.28	6.12	19/20 neg.

Note. Sorted by mean C_e , most negative first. [†]France $N = 1$ (semi-presidential system; single observation, interpret with caution). Italy: no $CT \geq 2$ observations. Binomial test: 19/20 countries negative, $p < 0.001$. Source: ParlGov (Döring & Manow, 2024).

4.5 Winner/loser decomposition

The aggregate mean C_e of -3.80 pp reported in Section 4.1 combines two structurally different groups. PM parties that retained the prime ministerial office after the election and those that lost it. For winners vs. zero, a two-sided test is applied because of a lack of directional assumption. PM parties that retained power after their re-election ($N = 53$, $df=52$) show a mean C_e of -0.33 pp. A one-sample t-test against zero yields $t(52) = -0.624$, $p = 0.535$ (two-sided), meaning that winning PM parties do not incur a statistically significant electoral cost. Notably, their mean is substantially higher than the N&P benchmark of -2.25 pp ($p < 0.001$, one-sided), suggesting that PM parties capable of retaining the prime ministerial office perform considerably better than the average government comprising all parties. In other words, the 52% of incumbent parties that win re-elections are not faced with a significant burden of incumbency. In contrast, PM parties that lost the prime ministerial office after their re-election bid ($N = 49$) show a mean C_e of -7.56 pp ($SD = 5.93$). A one-sample t-test against zero gives $t(48) = -8.93$, $p < 0.001$ (one-sided) and the mean lies far below the N&P benchmark ($p < 0.001$, one-

sided). These parties account for the major part of the aggregate PM cost.

On average, winners differ from losers by 7.23 pp. A two-sample t-test confirms that this difference is highly statistically significant ($t(100) = 7.37$, $p < 0.001$). This pattern is consistent with the mechanism of retrospective voting (Lewis-Beck, 1990, pp.3-8). Voters reward parties in power (PM) that have performed well enough to remain in office, while punishing those that have not. Therefore, what the aggregate C_e of -3.80 pp captures is the full distribution of PM re-elections. This is a distribution that includes both costless re-elections and severe vote share collapses. The PM incumbency cost found in this paper is not an equally distributed penalty on holding office. Rather, it is an asymmetric cost concentrated among governments that lose power.

It is important to make a critical note on causal interpretation of this result here. The winner/loser decomposition is not a clean causal analysis. By its design it is substantially endogenous. By definition a party with a sufficiently negative C_e is more likely to lose its re-election bid. Parties are classified as losers because their vote share declined enough to cost them their seats. Therefore, the difference in electoral cost between winners and losers cannot be used as evidence that losing causes a larger electoral cost. In fact, the direction of causality runs at least partially in the opposite direction. It is the large electoral cost that causes the loss of office. Therefore, the decomposition should be read descriptively, showing that there is a substantial amount of incumbents that do not suffer from the full incumbency cost rather than establishing a causal mechanism that distinguishes winners from losers.

4.6 Term-length decomposition

Another question that arises from the theory is whether the cost of PM incumbency increases or decreases with the length of incumbency. Stevenson (2002, p.157) identifies a positive correlation between cabinet duration and the cost of governing. The longer a party remains in office, the larger the cost of ruling it faces. The median-gap theory of Nannestad and Paldam (2000, pp.11–12) also argues in favour of an increasing incumbency disadvantage with longer tenure. As incumbents drift from their campaign positions over time, voter dissatisfaction accumulates. Both frameworks predict a harsher negative C_e when ruling longer. In other words, $CT \geq 3$ observations should show a more negative mean than $CT = 2$ observations (first term re-elections vs. those subsequent). However, the data covered in this paper does not support this argumentation. PM parties at their first re-election ($CT = 2$, $N = 44$) show a mean C_e of -3.68 pp ($SD = 6.45$, $SE = 0.97$), which is significant against zero at $t(43) = -3.79$, $p < 0.001$ (two-sided). PM parties at subsequent re-elections ($CT \geq 3$; $N = 58$) show a mean C_e of -3.89 pp ($SD = 5.90$; SE

= 0.78), which is also significant at $t(57) = -5.02$, $p < 0.001$ (two-sided). The difference between the two groups is 0.21 pp, which is negligible in magnitude and not statistically significant at conventional levels.

This null finding can be interpreted in two ways. First, within the 1990–2020 timeframe and the 20-country OECD sample, the structural cost of holding the prime minister office may dominate any accumulation effect of time in office. If concentrated electoral accountability (Powell & Whitten, 1993, p.398) is the primary driver of PM costs, accountability remains relatively constant across different re-elections, with voters targeting the PM incumbent party, regardless of whether it is in its second or fourth term. Second, it should be kept in mind that $CT \geq 3$ observations are, by definition, limited to PM parties that have already survived at least one re-election. This could introduce a selection bias that dampens the estimated $CT \geq 3$ cost. Despite this, the near-identical estimates across both groups suggest that the Stevenson time effect is not strongly observable within this specific sample.

5 Discussion

5.1 Interpretation of the PM cost of ruling

The primary finding is that PM parties lose 3.80 pp in re-elections where they hold the PM seat. This is 1.55 pp more negative than the N&P all-party benchmark. This finding is consistent with the theoretically treated concentrated accountability mechanism. In parliamentary coalition governments, voters can clearly identify the PM party as the main governing actor. This clarity of responsibility leads to voter dissatisfaction with government performance being primarily directed at the PM party and not distributed equally across all coalition members. The PM party therefore suffers from a disproportionately large share of the overall cost of ruling.

This interpretation is reinforced by the geographical patterns displayed in Table 3. Countries with large PM incumbency costs tend to have complex coalition politics and a dominant PM party. For example, Portugal, Norway and Spain all have mean PM costs more negative than the benchmark. In such systems, there is a large asymmetry between the PM party and its coalition partners, giving voters a clear target for electoral punishment (theoretically also for reward). In contrast, countries where the PM party's strength and visibility within the coalition is less pronounced, such as Austria and Luxembourg, show smaller costs, which is consistent with a more distributed blame and reward attribution.

The winner/loser decomposition from Section 4.5 allows for a further comment on the accountability of PM parties. The aggregate cost of -3.80 pp is composed of two different subgroup performances. While PM parties that retained office face virtually no statistically significant cost (-0.33 pp), those that lost power account for the majority of the penalty (-7.56 pp). This suggests that concentrated accountability does not act as a uniform burden on all PM incumbents, but more as a high-stakes accountability system for incumbent parties. PM parties that perform well during their reign face limited electoral punishment, while those that do not face severe consequences. Therefore, the mechanism is not a structural tax on office-holding and more a retrospective voting mechanism, which is consistent with Lewis-Beck's (1990, pp.3–8) economic voting framework. In this framework, voters punish incumbents only if performance is insufficient.

This finding also allows for a comment on the leadership premium hypothesis discussed in Section 2.3. If the leadership premium was significant, we would expect parties led by a highly visible prime minister to incur costs below the all-party benchmark. The prime minister's personal popularity would offset part of the cost of incumbency. However, the data does not support this prediction: On average, PM parties face a larger cost. This does not mean that there is no leadership premium. An extraordinarily popular prime

minister could in theory offset negative incumbency effects. The data, however, suggests that on average the leadership premium cannot help in overcoming the negative effects incumbent PM parties face.

Another interesting finding relates to tenure length. The data do not replicate the positive relationship between time in office and cost documented by Stevenson (2002, p.157) within this sample. One possible explanation is that the 1990–2020 timeframe and the specific OECD selection encompasses a period during which the mentioned concentrated accountability mechanisms overrule the effects of accumulation during reign. Another interpretation is that the results reflect a selection bias, where long-surviving PM parties are positively selected for electoral resilience. This could suppress the estimated cost suggested by theory at $CT \geq 3$. To disentangle these explanations would require a longer time horizon and a more sophisticated statistical approach.

5.2 Limitations

Macroeconomic controls: There is an absence of macroeconomic controls. The analysis measures the change in vote share C_e without controlling for the economic cycle. If PM incumbency periods from 1990 to 2020 were on average during worse-than-average economic conditions, the estimated PM cost would be influenced by controls that are not linked to incumbency. Conversely, if PM parties were able in timing elections during favourable economic periods, the estimate would be downward-biased. Disentangling the structural incumbency cost from the economic vote requires a more sophisticated statistical approach with growth and inflation controls, which is beyond the scope of this paper.

Sample independence: The t-test treats all 102 observations as independent, while in reality they are to some degree not independent within countries (with an average of 5.1 observations per country, $N = 20$).

Temporal comparability: The Nannestad and Paldam (2000) benchmark of -2.25 pp is from a sample of 282 elections between 1948 and 1997. This paper covers the period from 1990 to 2020, which only partially overlaps with the benchmark sample. Therefore, part of the difference between the observed PM cost effect of -3.80 pp and the benchmark of -2.25 pp may reflect an increase in incumbency penalties for all governing parties during this more recent period, rather than a PM-specific surplus cost. Without an all-party benchmark calculated over the same 1990–2020 period, the 1.55 percentage point difference cannot be exclusively attributed to the PM position.

France: France provides only one observation from an unusual context: the semi-presidential system of the Fifth Republic, in which the executive authority is shared between the president and the prime minister (Messaoudi & Belaghouati, 2025, pp.2997–2998). The 1997 observation (the RPR/UDF combination government) is included for the completeness, but it should be interpreted with caution. Still, excluding France shifts the mean from -3.80 to -3.72 percentage points, leaving all results and hypotheses confirmations unchanged.

Italy: Italy actually has no $CT \geq 2$ observations which is a genuine feature of Italian politics. There is extreme volatility in PM seat occupancy in Italy. This absence is not a data problem. Every Italian general election between 1990 and 2020 actually produced a new PM party in ParlGov’s cabinet data. As a result, Italy is excluded from the count of OECD countries due to providing no incumbency observations ($N = 20$, not 21).

6 Conclusion

This paper analysed whether prime ministerial parties face different electoral costs than other governing parties. The answer is clear: Yes, they do. The observed difference is a cost larger in magnitude, not a smaller one. Across 102 prime ministerial incumbent elections in 20 OECD parliamentary democracies between 1990 and 2020, incumbent parties lost an average of 3.80 percentage points of their vote share per election. This figure is significantly negative ($p < 0.001$) and also significantly more negative than the N&P all-party benchmark of -2.25 percentage points ($p = 0.006$). This pattern is consistent in geography, holding true in 19 out of 20 sample countries. This confirms that the PM cost premium is not an artefact of some exceptional outliers, it is a structural feature of modern democratic parliamentary systems.

These findings do not support mechanisms of prime ministerial insulation. The personal leadership premium is apparently not large enough to reduce the average cost of the PM party below the all-party benchmark. Instead, the findings support the concentrated accountability mechanism, as voters can easily identify the PM party as the primary government actor. This means that electoral punishment is stronger and more reliable for PM parties than for junior coalition partners. The argument provided is that junior partners can more credibly distance themselves from (negative) government policies. Consequently, in the analysed sample, the party holding the most powerful government office is also faced with a larger electoral cost of holding it.

The N&P benchmark of -2.25 pp serves as a useful reference point, however, it groups together parties that hold very different positions within the government hierarchy. Empirical evidence from this paper suggests that the cost of ruling framework cannot be applied to all ruling parties homogeneously. Disaggregating the benchmark by government role would be more informative. PM parties carry a cost of approximately -3.80 pp, while the all-party average of -2.25 pp likely reflects a combination of a higher cost associated with PM parties and a lower cost associated with junior partners. A valuable extension to these findings would be introducing macroeconomic controls, including growth, inflation and unemployment. With macroeconomic controls, the structural PM cost could be separated from the economic vote. This would answer the question of if and to what extent the PM premium persists after taking economic conditions into account. Secondly, extending the analysis to longer time horizons and a broader set of countries would test whether the results can be generalised beyond the 1990–2020 OECD sample and would allow for a more direct comparison to the N&P benchmark (Nannestad & Paldam, 2000).

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Data Sources

- Döring, H., & Manow, P. (2024). *ParlGov 2024 Release [Dataset]*. <https://doi.org/10.7910/DVN/2VZ5ZC>

List of Aids

Aid / Tool	Purpose
BibLaTeX (Backend Biber)	Source management, bibliography
Claude Code (Anthropic)	Writing, editing, debugging and commenting of Python code (vibe coding)
Google Translate / DeepL	Translation of single words (German to English)
Grammarly AI; DeepL Write	Corrections regarding grammar and readability of the text
Numiqo (YouTube)	Self-teaching and understanding of statistics (www.youtube.com/@numiqo)
Perplexity AI (LLM's used: Sonar, Claude Sonnet 4.6, Nemotron 3 Super)	Aid in structuring of the paper, aid in brainstorming and idea generation, aid in creating the paper layout in LaTeX, aid in creating figures in LaTeX, aid in self-teaching and comprehension of data analysis / statistics
Python 3.11. with: pandas, SciPy, statsmodels and matplotlib	Data analysis and statistical tests
Swisscovery, Google Scholar	Search tool for academic sources

Affected parts:	All aids have been used globally, encompassing the full paper
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Appendix

Python replication code for statistical analysis: Fellner, N. (2026). OECD PM incumbency-effects [Python code]. GitHub. https://github.com/nkf-university/atp_econ.git